

CRISTOFF STAN

Senior Software Engineer

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PROFESSIONAL SUMMARY

Seasoned Senior AI/Software Engineer with 9+ years of hands-on experience delivering production-grade AI/ML systems using Python, LLMs, NLP, and cloud platforms like AWS and GCP. Adept in building and deploying GenAI and ML models using frameworks like LangChain, LlamaIndex, TensorFlow, Torch, and Scikit-learn. Skilled in designing robust RAG pipelines, prompt engineering, embeddings, and vector databases. Proven track record of developing scalable applications using microservices, Docker, and Kubernetes. Passionate about clean architecture, secure coding, and optimizing performance in AI systems under real-world constraints in regulated domains.

TECHNICAL SKILLS

- **Programming:** Python, JavaScript, TypeScript, Java
- **ML/AI/NLP:** LangChain, LlamaIndex, ReAct, Scikit-learn, TensorFlow, PyTorch, Spacy, XGBoost, Hugging Face, OpenAI, Claude, Amazon Titan, Meta LLaMA, Mistral
- **GenAI Tools:** Prompt Engineering, RAG, CoT, ToT, Guardrails, Embeddings, Output Parsers
- **Cloud Platforms:** AWS (SageMaker, Lambda, Bedrock, SQS, CDK), GCP, Docker, Kubernetes
- **Databases:** PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Pinecone, FAISS
- **DevOps & Tools:** Git, CI/CD, Jenkins, Terraform, Airflow, Kafka, REST APIs, Flask, FastAPI
- **Data & Infra:** Pandas, NumPy, Scipy, Apache Spark, SQL, NoSQL
- **Security & Privacy:** Data privacy principles, encryption, secure coding

WORK EXPERIENCE

Senior Software Engineer

Sensidev

September 2021 – December 2024

Timișoara, Romania (Remote)

- Designed and deployed production-level NLP models using Hugging Face Transformers and LangChain for semantic search and question answering in medical documents.
- Built multi-stage RAG pipelines using FAISS vector DB and OpenAI models, achieving 88% accuracy improvement on entity recognition from unstructured text.
- Led architecture of an LLM-powered chatbot integrated with internal tools, using ReAct-style agents and custom tools over AWS Bedrock and Pinecone.
- Integrated SageMaker endpoints with FastAPI microservices to enable scalable ML inference with autoscaling and latency monitoring via CloudWatch.

- Developed robust prompt engineering frameworks to support GenAI applications for contract summarization and clause detection in regulatory documents.
- Implemented feature selection pipelines using XGBoost and SHAP for customer behavior prediction, improving retention models by over 27%.
- Applied guardrails to sensitive GenAI workflows to enforce deterministic responses using regex, JSON schema validators, and fallback mechanisms.
- Mentored a team of 6 engineers on Python-based AI microservices, CI/CD best practices, and containerization with Docker and Kubernetes.

Software Engineer

Syscom Digital

October 2019 – August 2021

Bucharest, Romania (Remote)

- Developed ML pipelines for retail customer segmentation using unsupervised clustering, feature engineering, and imbalance resampling techniques.
- Built document parsing services with Spacy and OCR integrations that handled over 10M PDFs per month in a distributed, containerized pipeline.
- Designed secure tokenized data processing with field-level encryption and privacy auditing as per GDPR requirements.
- Applied AWS Lambda and SQS to orchestrate event-driven pipelines for on-demand ML scoring services.
- Created custom evaluation dashboards with SQL/NoSQL metrics reporting to monitor precision, recall, F1-scores, and latency for deployed AI models.
- Implemented caching layers using Redis and optimized PostgreSQL queries for analytics APIs that reduced response times by 60%.
- Collaborated with frontend and DevOps teams to define interfaces, infrastructure, and testing strategies for AI-integrated dashboards.
- Documented APIs and architecture using OpenAPI and Confluence, streamlining onboarding and reducing support tickets by 40%.

Software Developer

Soft Build

June 2016 – July 2019

Timișoara, Romania (Remote)

- Engineered backend AI services using Flask, Torch, and pandas to handle structured and unstructured analytics for finance and HR platforms.
- Implemented text classification models using logistic regression, SVMs, and TF-IDF pipelines to tag and categorize multilingual documents.
- Managed SQL/NoSQL hybrid data layers optimized for distributed storage and real-time analytics on MongoDB and PostgreSQL.
- Introduced unit testing for ML models using PyTest and coverage enforcement in CI pipelines, increasing model testing coverage from 40% to 95%.
- Refactored legacy pipelines using Apache Airflow for ETL workflows, improving scheduling and reducing operational overhead.
- Set up containerized staging environments using Docker Compose, reducing QA feedback cycles by 45%.
- Applied model interpretability techniques like LIME to enhance stakeholder trust in ML-based recommendations.
- Conducted monthly knowledge-sharing sessions on applied NLP, leading to adoption of new standards across multiple teams.

Software Developer
Decima Digital
November 2015 – April 2016
Tallinn, Estonia

- Created RESTful services for NLP-based document matching using cosine similarity and vector space models.
- Wrote reusable Python libraries for data normalization and tokenization, reducing code duplication by 60% across teams.
- Performed experimentation with early Word2Vec and GloVe embeddings to enrich semantic search tools.
- Built rule-based NLP pipelines for entity extraction in Estonian legal texts, deployed as microservices behind API Gateway.
- Helped migrate legacy scripts into well-documented Python modules and Docker containers to support CI integration.
- Conducted internal tech sessions on NLP pipelines and evaluated performance trade-offs for regex vs ML-based approaches.
- Benchmarked accuracy across TF-IDF, LSI, and early BERT-based approaches on internal datasets.
- Developed automated test scripts and documented behavior for continuous integration validation.

Software Developer Intern
Borna Technology
September 2014 – August 2015
Tallinn, Estonia (On-Site)

- Developed Python-based utilities for log analysis and anomaly detection in system telemetry data from industrial sensors.
- Conducted time-series forecasting using ARIMA and Holt-Winters for predictive maintenance dashboards.
- Created internal data visualization tools using Plotly and Matplotlib for non-technical stakeholders.
- Optimized SQLite queries used in field applications, improving report generation time by 80%.
- Built early prototypes for classification tasks using decision trees and logistic regression for parts classification.
- Learned industry standards on version control, Agile workflows, and debugging in production environments.
- Shadowed senior engineers and participated in pair programming and code review sessions.
- Delivered internal presentation on intro to scikit-learn and basic ML workflows, gaining cross-team recognition.

EDUCATION

Tallinn University

Bachelor's Degree in Computer Science,
2010 – 2014

CERTIFICATIONS

- ❖ AWS Certified Machine Learning – Specialty
- ❖ TensorFlow Developer Certificate – DeepLearning.AI

- ❖ Prompt Engineering for LLMs – DeepLearning.AI
 - ❖ Data Science and GenAI with Python – Coursera
 - ❖ Kubernetes for Developers – Udacity
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PROJECTS

SmartContractGenAI

- Designed a GenAI pipeline that parses legal contracts, summarizes key clauses, flags anomalies, and generates counterparty suggestions using LangChain + OpenAI GPT-4.

VectorSearch Platform

- Built a production-scale RAG framework integrating Pinecone, LlamaIndex, and OpenAI with prompt chaining to deliver contextual Q&A from enterprise documents.

Document Intelligence Hub

- Led development of a document parsing and classification system using Spacy and custom LSTM models, processing over 50M PDFs with 92% accuracy.

Customer360 AI Engine

- Created an ML engine to segment, classify, and score customers using behavior-based clustering and predictive scoring across multiple touchpoints.